

Presentation 5:

- Links to Presentation:
 - Presentation (Slides 12-14)
 - https://github.com/arvindkrishna87/STAT390_LegalAid_Fall2025/blob/main/Presentations/Call%20outcome%20analysis/Call_Classification_10_Dec_Sana.pdf
 - Video sent through email.
- What did you do?
 - This time, with my team, we verified and finalized the spreadsheet for call classification and did some analysis on our results.
- How does this help the project?
 - Gives a good top-down review of all calls in the dataset, especially for stakeholders who are beginning to understand the work and teams that are working on implementing the changes for the automated call system.
- Issues/Attempts to Resolve/ Resolved Issues:
 - No issues faced
- References:
 - Our team all used the same spreadsheet and validated the work we did within it.

Presentation 4:

- Links to presentation(s) and code(s) on GitHub:
 - Code
https://github.com/arvindkrishna87/STAT390_LegalAid_Fall2025/blob/main/Code/Call%20outcome%20analysis/MainMenuAbandoned_19Nov_Sana.ipynb
 - My Slides:
https://github.com/arvindkrishna87/STAT390_LegalAid_Fall2025/blob/main/Presentations/Call%20outcome%20analysis/MainMenuAbandonment_19Nov_Sana.pdf
- What did you do?
 - This time, I continued the work from last week's presentation and focused on the main menu and abandonment within it. Alongside that, I also filled out the CAR abandonment spreadsheet.
- How does this help the project?
 - It helps the project because I'm able to make recommendations on where customers leave calls (with the understanding that they may have not received the legal assistance they came for). I'm also able to see specific common routes that customers take.
- Issues faced (if any):
 - Separating Telephony EP versus Menu. The Main Menu is part of the Main Number EP but so is the language selection transfer before it.
- Attempts to resolve issues (if any)
 - I filtered those values out, but I want to be sure of my logic.
- Issues resolved (if any)
 - Again, I was able to filter the values out, but I'm also wondering about calls entering other EPs immediately after the Main Menu/Main Number EP?
- References (Mention if you built up on someone else's work)
 - The team and I used the same data sets (CAR after March 16th) and imported it all the same ways so we had a consistent number of unique calls.
- Next Steps:
 - Understanding any overlap that may exist between different menus/EPs and how,

if at all, the values we are getting need to be adjusted. Looking at logic behind transfers and why the abandoned rate for front desk transfer 3 is high. Looking at connecting menus (Language selection/Appointment menus)

Presentation 3:

- Links to presentation(s) and code(s) on GitHub:
 - Video Presentation:
https://github.com/arvindkrishna87/STAT390_LegalAid_Fall2025/blob/main/Presentations/Call%20outcome%20analysis/Abandonment_Nov5_sana_meera_adam_marcos.mov
 - My Slides:
https://github.com/arvindkrishna87/STAT390_LegalAid_Fall2025/blob/main/Presentations/Call%20outcome%20analysis/HousingAbandonment_5Nov_Sana.pdf
 - My Code:
https://github.com/arvindkrishna87/STAT390_LegalAid_Fall2025/blob/main/Code/Call%20outcome%20analysis/HousingMenuAbandoned_4Nov_Sana.ipynb
- What did you do?
 - This time, alongside the rest of the team, I took a look at the housing menu and abandonment within this specific menu. This involved not only looking at directly after which activity the customer left but also whether they navigated a specific route before leaving.
- How does this help the project?
 - This helps the project because I'm able to make recommendations on where customers leave calls (with the understanding that they may have not received the legal assistance they came for). I'm also able to see specific common routes that customers take (for example, customers immediately entering the tenant menu then leaving)
- Issues faced (if any):
 - Narrowing down the not captured percentage of abandoned calls. There are many calls in which a customer leaves but an activity is not captured in the row preceding it. What does that mean?
- Attempts to resolve issues (if any)
 - I looked at two rows preceding it to count the number of calls in which the actual

reason of leaving was within housing (many were found under housing or tenant menus)

- Issues resolved (if any)
 - While what I did lowered the uncaptured percentage, the number is not exactly zero and the ones I did manage to capture (by examining two rows above for the activity name, I have not been able to track their full route → did they leave only after reaching the tenant or did they take several steps back and then left?) I want to match my analysis that I did earlier.
- References (Mention if you built up on someone else's work)
 - The team and I used the same data sets (CAR after March 16th) and imported it all the same ways so we had a consistent number of unique calls.
- Next Steps:
 - I want to further understand the issues I had mentioned earlier, and focus on other menus/submenus as well.

Presentation 2:

- Links to presentation(s) and code(s) on GitHub:
 - Presentation slides (slides 6-9):
https://github.com/arvindkrishna87/STAT390_LegalAid_Fall2025/blob/main/Presentations/Call%20outcome%20analysis/CustomerLeft_20Oct_Sana.pdf
 - Code:
https://github.com/arvindkrishna87/STAT390_LegalAid_Fall2025/blob/main/Code/Verbiage%20analysis/CustomerLeftScenarios_20Oct_Sana.ipynb
- What did you do?
 - This time, alongside Meera, we looked at specific scenarios in which the customers were leaving the call in the CAR data. I looked at call journeys (going through many menus just to leave in the end) and other things like customers waiting too long in queues. I also paired some of the specific examples with the verbiage of the prompt to better contextualize why some customers may have left at this specific menu (such as looking at customers going back to a previous menu before leaving → implies that they didn't receive assistance in some capacity at the last menu).
- How does this help the project?
 - This helps the project because I'm able to make recommendations on where customers leave calls (with the understanding that they may have not received the legal assistance they came for). For example, the queue wait times are common reasons as to why customers leave, but there is always the closed queue option, so can Legal Aid be more liberal with when they decide to close the queues?
- Issues faced (if any):
 - I struggled with understanding the long wait times at the Farmworker menu that ended up with the customer leaving the call. There wasn't much clarity on whether they were being transferred or something else.
- Attempts to resolve issues (if any)
 - I looked closer at the verbiage prompt and the flow charts to see what possible

paths the customer could go to after that menu. There is the direction to the business hours as well as the voicemail transfer (but that is not logged in the CAR).

- Issues resolved (if any)
- References (Mention if you built up on someone else's work)
 - I worked with Meera on this, so to maintain consistency, I used her code in developing the time difference column and organizing call IDs in a more simple way (numbered) vs have the longer session ID names.
- Next Steps:
 - I want to further understand some of the pathways (specifically the farmer menu) to continue looking at why the customers left as well as ways to better format the legal menus which are some of the most common places that customers leave.
 - I also want to work with Meera on her analysis with the Chatbot implementation.

Presentation 1:

- Links to presentation(s) and code(s) on GitHub:
 - https://github.com/arvindkrishna87/STAT390_LegalAid_Fall2025/blob/main/Code/Resource%20allocation%20analysis/PhoneLineVariabilityAndLanguageAllCalls_Oct6_Sana.ipynb
 - First Link is where main code for language distribution and phone line data is. This was after figuring out two phone lines that led to English/Spanish main menus.
 - https://github.com/arvindkrishna87/STAT390_LegalAid_Fall2025/blob/main/Code/Resource%20allocation%20analysis/PhoneLineVariability_Oct6_Sana.ipynb
 - The links above are to my code for resource allocation. There's some code and analysis that isn't mentioned in the presentation because there was not much that I learned from it.
 - https://github.com/arvindkrishna87/STAT390_LegalAid_Fall2025/blob/main/Presentations/Resource%20allocation%20analysis/Resource_Allocation_Presentation_1_Oct7_Sana.pdf
 - This is the link to the slides for presentation 1. My updates are on slides 12-17.
- What did you do?
 - I focused on presentation one to get a better understanding of the data queues and the routes callers make to get to their destination. My two main focuses were on phone line distribution and language choices. I was hoping to see if there were some large discrepancies in which lines were called more than others, especially during different times of the year/day/week. I was also hoping to see language distributions, however, I am still in the process of understanding the phone logs to identify which callers are selecting Spanish as their language or waiting for an interpreter at the language menu.
 - After Friday Mentor Meeting, was able to look at duration of calls between Spanish and English calls by filtering by inbound calls and grouping by correlation ID and finding averages of calls (Spanish and English)

Spanish calls were by average 78 seconds longer. Why?

- How does it help the project?
 - This should help the project because it informs us and Legal Aid of how the current system is being used by customers and if their intended audience/clients are being served. (Spanish and other language-speakers). It can also help improve the menus by possibly finding redundancies (such as the choice of where to place the language menu)
- Issues faced (if any):
 - My main issue with looking at the language chosen for each call was finding the 'activity' or prompt where customers selected the language. According to the verbiage spreadsheet, there are WCC prompts in both Spanish and English. However I'm never seeing a single instance of any prompt that is in Spanish in both the CAR and All Call data. The only example is when customers select the staff directory option from the main menu in Spanish. At that point they've already selected Spanish from the first language selection menu, so why is the main menu not being shown as main_menu_spanish?
 - I'm also interested in understanding which phone numbers are being used. I'm referencing this spreadsheet to understand which phone number is to what department/organization, however I'm not seeing any of these numbers in the data? I'd like to better understand how to track this.
- Attempts to resolve issues (if any)
 - I attempted to resolve this issue by searching for different examples of Spanish being mentioned or any indication of what the caller selected in the language menu but I'm still confused about this.
 - Update: After Friday Mentor Meeting, was able to find some indication of Spanish/English calls in the All Cars Dataset. However, still would like some indication of more examples of Spanish queues (did people switch over from Spanish to English or vice versa later on in their call?)
- Issues resolved (if any)
 - N/A
- References (Mention if you built up on someone else's work)
 - N/A
- Next Steps:
 - I'm interested in looking at other ways of identifying English/Spanish calls, especially to look at different switches during a singular call. Maybe a Spanish speaker who requests Spanish at a later time of the call? Or similar instances. I also am interested in looking at other ways to compare the English and Spanish calls outside of the duration of calls.